



TONG BAO

Nationality: China

Institute: I.N.A.F - Osservatorio Astronomico di Brera, Italy

+86-18801589508

tong.bao@inaf.it

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EMPLOYMENT

- I.N.A.F - Osservatorio Astronomico di Brera** Oct 2024–now
Postdoc

EDUCATION

- Nanjing University** Sep 2019 – June 2024
Ph.D., Astrophysics Advisor: Prof. Zhiyuan Li
Thesis: The Study of X-ray Periodic Variability in the Galactic Bulge and Globular Clusters
- Nanjing University** Sep 2015 – June 2019
B.S., Astronomy
Thesis: Application of Neural Network Algorithm in X-ray Point Source Detection

ACADEMIC EXPERIENCE AND COLLABORATIONS

- Member of the HotMilk group at INAF** June 2024–now
investigating the hot phase of the Milky Way, the so called Galactic corona. Remote between June 2024 to Sep 2024
- Summer research project** July 2019-Sep 2019
Application of machine learning in Astronomy
- Columbia University** April 2023 - April 2023, New York
Periodic X-ray sources and exotic binaries in dense environment Host: Prof. Charles J. Hailey & Dr. Kaya Mori
- Columbia University** June 2018 - Sep 2018, New York
Research Assistant-NuSTAR hard X-ray sources in the Galactic center: HMXB candidates Host: Prof. Charles J. Hailey

RESEARCH INTERESTS

- Periodic/Aperiodic variabilities in X-ray binaries
- Stellar populations and dynamics in Globular clusters and the Galactic center
- Quasi-periodic oscillations/eruptions of AGNs
- Exotic X-ray binaries (double degenerate stars)

PROPOSALS AND AWARDS

- NuSTAR, Cycle 10; General Observing** 2024
PI: Shifra Mandel. Co-Is: Tong Bao, et al
- XMM-Newton, Cycle 24; General Observing** 2024
PI: Samaresh Mondal. Co-Is: Tong Bao, et al
- XMM-Newton, Cycle 24; General Observing** 2024
PI: Heshou Zhang. Co-Is: Tong Bao, et al
- President's Exceptional Scholarship** Nanjing university, 2019
Annually awarded to the top 10% doctoral student.
- Scholarship of National Astronomical Observatories** Chinese Academy of Sciences, 2017
Awarded to undergraduate students with strong research experience.
- People's Scholarship** Nanjing university, 2017
Second class, annually awarded to the top 10% undergraduate students.
- People's Scholarship** Nanjing university, 2016
Third class, annually awarded to the top 20% undergraduate students.

ORAL PRESENTATIONS

- **X-ray Universe 2026** Elche, Spain
X-ray-luminous late-type giants: an overlooked population contribute to Galactic ridge iron line emission 2026
- **NewAthena Workshop SWG4 2026** Barcelona, Spain
A comprehensive census of X-ray periodic sources in dense star cluster: from Chandra to NewAthena 2026
- **Invited talk, INAF/IASF-MI** Milan, Italy
Periodic X-ray sources: diagnosing the nature and evolution for exotic binaries in dense environment 2025
- **Como Lake center for AstroPhysics Inaugural Workshop** Como, Italy
Unveiling the Soft X-ray Source Population of the Inner Galactic Disk with XMM-Newton 2025
- **Invited talk, Department of Physics, Columbia University** New York, USA
Periodic X-ray sources: diagnosing the nature and evolution for exotic binaries in dense environment 2023
- **eROSITA Scientific Exploitation Workshop** Suzhou, China
Exploring the Outskirts of Globular Clusters with eROSITA 2023
- **Research on the Variations of Active Galactic Nuclei** Xiamen, China
Searching for Quasi-periodic Oscillations in Active Galactic Nuclei of the Chandra Deep Field South 2023
- **Structure, Formation, and Evolution of Star Clusters** Zhuhai, China
Diagnosing dynamical formation and evolution of cataclysmic variables in globular clusters 2023
- **The 60th Anniversary of X-Ray Astronomy** Beijing, China
Searching for Quasi-periodic Oscillations in Active Galactic Nuclei of the Chandra Deep Field South 2022
- **2021-2022 Academic Annual Meeting of Jiangsu Astronomical Society** Nanjing, China
Periodic X-ray Sources in the Galactic Center/Bulge – Application of the Gregory-Loredo Algorithm 2022
- **Chandra Data Science: Novel Methods in Computing and Statistics for X-ray Astronomy** Online
Searching for Quasi-periodic Oscillations of Active Galactic Nuclei in the Chandra Deep Field South 2021
- **The 4th X-ray binary Multi-Band research Symposium** Kunming, China
Diagnosing dynamical origin of exotic binaries in the massive globular cluster 47 Tucanae 2021
- **Chandra Frontiers in Time-Domain Science** Online
Periodic X-ray Sources in the Galactic Center/Bulge – Application of the Gregory-Loredo Algorithm 2020
- **The 3rd X-ray binary Multi-Band research Symposium** Jinan, China
Periodic X-ray Sources in dense environment 2020

TEACHING AND MENTORING

- **Teaching Assistant** Nanjing university
Course: "An Introduction to X-ray Astronomy" 2020, 2021, 2022
– X-ray data processing training project (https://github.com/baotong6/xray_astronomy), teaching students how to reduce the X-ray data, analyze the spectrum and light curves of CVs, X-ray binaries, ULXs, AGNs, etc.
- **Students mentoring** Nanjing university
2021-2022
– Teodora Darie, Erasmus+ traineeships: “The X-ray spectral properties of different types of main-sequence stars”
– Jiachang Zhang, Undergraduate thesis: “Searching for periodic X-ray binaries in M31 bulge”
– Jiachang Zhang & Yufan Zhou, Research projects during semesters, “Detecting Emission Line Point Sources in M31 bulge”

PUBLICATIONS (FULL LIBRARY ACCORDING TO ADS)

- [1] **Bao. T.**, Ponti, G., et al., 2026, A&A, 710, A243, “X-ray-luminous late-type giants: An overlooked population contributing to the Galactic ridge iron line emission”
- [2] **Bao. T.**, Ponti, G., et al., 2025, A&A, 704, A107, “Unveiling the soft X-ray source population toward the inner Galactic disk with XMM-Newton”
- [3] **Bao. T.**, Li, Z., Cheng, Z., Belloni, Diogo. 2024, MNRAS, 527, 7173, “A Chandra Survey of Milky Way Globular Clusters – IV. Periodic X-ray sources”

- [4] **Bao, T.**, Li, Z., & Cheng, Z. 2023, MNRAS, 521, 4257, “Periodic X-ray sources in the massive globular cluster 47 Tucanae: Evidence for dynamically formed cataclysmic variables”
- [5] **Bao, T.**, & Li, Z. 2022, MNRAS, 509, 3504, “Searching for quasi-periodic oscillations in active galactic nuclei of the Chandra Deep Field South”
- [6] **Bao, T.**, & Li, Z. 2020, MNRAS, 498, 3513, “Periodic X-ray sources in the Galactic bulge: application of the Gregory-Loredo algorithm”
- [7] Zhang. J, **Bao. T.**, Li, Z., 2024, “A Chandra Search for Periodic X-ray Sources in the Bulge of M31”, 2024, MNRAS, 530, 2096.
- [8] Wang Z., Rea N, **Bao, T.**, et al., 2025, Nature, 642, 583, Detection of X-ray emission from a bright long-period radio transient
- [9] Mondal, S., Ponti, G., **Bao, T.**, et al., 2025, "A sample of ionised Fe line-emitting X-ray sources in the inner Galactic disc", 2025, A&A, 698, 206.
- [10] Mondal, S., Ponti, G., **Bao, T.**, et al., 2024, "Periodicity from X-ray sources within the inner Galactic disk", 2024, A&A, 686, 125.
- [11] Zhang. J, Li, Z., Hua, Z., **Bao. T.**, 2026, “Probing periodic and aperiodic variability of X-ray sources in M31, M81, and Centaurus A with Chandra”, 2026, MNRAS, 546, 2237.
- [12] Mondal, S., Ponti, G., Filor, L., **Bao, T.**, et al., 2024, A&A, 689, A172, "XMM-Newton and NuSTAR discovery of a likely IP candidate XMMU J173029.8-330920 in the Galactic Disk"
- [13] Ma. Q, Zhang. J, Gu. W, Li, Z, Weng. S, **Bao, T.**, 2025, The Innovation, 101181, "The first extragalactic ultra-compact X-ray binary"

PUBLIC OUTREACH

- **YOUNG, Orienta il tuo futuro** *Como*
Volunteer 2025
- **“Astronomical Night for the Public” Activity of Zuo Dijiang Observatory** *Nanjing*
Volunteer 2021, 2022, 2023
- **Summer Camp for High School Students at the School of Astronomy and Space Science** *Nanjing*
Volunteer 2022
- **Undergraduate Summer Camp at the School of Astronomy and Space Science** *Nanjing*
Volunteer 2021
- **Zuo Dijiang Observatory Open Day Activities** *Nanjing*
Lecture Speaker 2020